Tô Gia Bảo

Data Engineer

Thu Duc, Ho Chi Minh City, Vietnam

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Summary

Data Engineer in digital banking, focused on reliable ETL pipelines, data quality, and analytics with Python and SQL. Hands-on with automated testing, migration reconciliation, PII-aware processing, and pipeline observability across AWS and dbt. Delivered measurable S3 storage savings and self-service analytics backed by governed metadata.

Experience

Timo Digital Bank by BVBank — *Ho Chi Minh City, Vietnam*

Data Engineer — *Mar. 2025 – Present*

- Implemented MariaDB sync workflows with Amazon EMR and AWS Glue, applying PII masking between landing and curated layers and dbt transformations for analytics-ready data.
- Used dbt tests, Great Expectations, and pytest for automated data validation and regression testing, including query contracts, SQL compilation, and dbt/Lightdash metadata integration checks.
- Optimized one S3 table with Parquet ZSTD-3 compression and file compaction, saving 38.4 TB (45%) and reducing object count by 80% across 56.8 billion rows (2023 to Sep. 2026). Audited 1,360 partitions using migration-recorded row counts and fingerprints, reconciled with Glue and S3 metadata.
- Developed a FastMCP server powering conversational analytics in Slack, grounded in dbt docs, Lightdash dashboards, and semantic-layer metadata for self-service data discovery.
- Automated MongoDB backup validation and general ledger lookups with Python and Airflow, reducing manual checks.
- Monitored Airflow, Athena, and S3 health through dbt and Lightdash dashboards; migrated Timo Metrics from Loki/Grafana to SigNoz to centralize logs and metrics for SLA tracking and troubleshooting.
- Delivered Tracking247 to sync Customer-provided shipment IDs and status updates into MongoDB via Airflow and Streamlit.

Projects

Mini Lakehouse — *GitHub*

Lakehouse & Platform Engineering — *Jul. 2026 – Present*

- Built a contract-driven AWS lakehouse on S3, Iceberg, Glue, Athena, and EMR Serverless, with Airflow and platform services self-hosted on OCI and infrastructure managed through Terraform.
- Delivered deterministic batch and streaming pipelines for GitHub activity, arXiv research, and SSI market data, with dbt/Lightdash analytics, Modal GPU OCR, and Netdata observability exposed through Cloudflare Zero Trust without public host ingress.

VN Market Pulse — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese research pipeline with multi-source search, deduplication, and reranking to generate source-backed posts.
- Used PydanticAI for typed output and validation, with retrieval budgets, retries, usage tracking, and CLI/Streamlit interfaces.

Technical Skills

Programming	Python, SQL, C++
Processing & Orchestration	Databricks, Spark/PySpark, Airflow, dbt, Polars
Data Quality & Testing	dbt tests, Great Expectations, pytest; data reconciliation, contract and regression testing
Cloud Platforms	AWS (S3, Athena, EMR, Glue)
Azure Familiarity	Azure Data Factory (ADF), Microsoft Fabric, OneLake
Lakehouse & Databases	Apache Iceberg, PostgreSQL, MariaDB, MongoDB
Infrastructure & DevOps	Terraform, Docker, Git
Observability	SigNoz, Netdata, Grafana, Loki
AI & Analytics	Lightdash, Streamlit, Model Context Protocol (MCP), FastMCP, PydanticAI

Education

University of Science, VNU-HCM (HCMUS) — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

Publication

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval — *SOICT*

Co-author — *2025*

Languages

English TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350